Adaptive Fabric Surface Defect Detection and Localization Using Classical Texture Descriptors

Team TEXTURA

Weekly Project Progress Report – Week 7

Course: Digital Image Processing

Repository link:

https://github.com/AditiKapadiya/DIP_2026_Project17_Group8.git

Team Members

Aditi Kapadiya	AU2440139
Disha Gadhiya	AU2440217
Khushi Panchal	AU2440002
Harshita Dhanwani	AU2440083

Work Done

- Constructed one evaluation notebook to evaluate six methods in exactly the same way, which are: LBP, GLCM, Gabor, fixed one scale Fourier, fixed multi-scale Fourier and adaptive multi-scale Fourier.

- Maintained a balanced distribution of the data. Defect patches are divided into a development set (DEV) and a test set (TEST) by source image and no image will be included in both sets. Images that are free of defects are divided into three sections for each fabric: REF (learn normal texture), CAL (set thresholds) and NTEST (count false alarms). The TEST set has not been used previously.
All methods used were trained the same. Both learning methods require normal texture as input, which is obtained from REF, and both require a score between 0 and 1 to scale the results, which is supplied by CAL, and both require a threshold to set the percentage of defect-free patches that are reported, which is set at 5% in both methods.
Applied the same steps to the mask for each method: threshold, optional region growing, cleaning, and small blob removal. For each method, Growth is selected for DEV only.
Incorporated pixel-level metrics: IoU, precision, recall and F1 (mean per patch, mean over all patches), plus average precision and AUROC on the deviation map that are not dependent on the threshold.

- Added defect-level metrics. A defect is scored as detected (well covered) if any predicted pixel overlaps it, otherwise it is not. This will also be divided into small, medium and large defects and false regions counted per patch.

- Added image-level metrics. We report precision, recall, F1, false-alarm rate, AUROC, and average precision based on the confidence score and each patch is labelled defective or defect-free.
These are bootstrap confidence intervals for source images, paired comparisons of adaptive vs fixed and best Fourier method vs best base line.

- Rechecked the confidence score whether it is calibrated on defect free patches, whether it is following the segmentation accuracy.
Stored results per patch and per defect, including fabric, source image size and defect size, to be analysed next week.

- Tested the notebook. It executed end to end on synthetic data in both quick mode, full mode and final-test mode and the metric and the bootstrap functions were compared to a hand-calculated example. The use of gabor filtering was made approximately 27 times faster without any loss of efficiency.

Results

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============================================================
PIXEL-LEVEL RESULTS (DEV, 152 defect patches)
============================================================
                          Mean IoU  Mean F1  Pooled IoU  Pooled P  Pooled R  Pooled F1     AP  Macro AP  AUROC
LBP                          0.000    0.000       0.000     0.000     0.000      0.000  0.031     0.102  0.504
GLCM                         0.004    0.007       0.009     0.014     0.029      0.019  0.030     0.099  0.501
Gabor                        0.012    0.020       0.017     0.024     0.052      0.033  0.093     0.055  0.822
FFT fixed single-scale       0.023    0.035       0.011     0.016     0.034      0.021  0.035     0.116  0.551
FFT fixed multi-scale        0.032    0.052       0.013     0.018     0.043      0.025  0.036     0.209  0.552
FFT adaptive multi-scale     0.074    0.104       0.076     0.089     0.340      0.141  0.080     0.084  0.797

Defect pixel fraction (AP chance level): 0.0328
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============================================================
DEFECT-LEVEL RESULTS (163 true defects)
============================================================
                        Detected (any)  Covered (>=50%)  Detected small  Detected medium  Detected large  False regions / defect patch  False regions / normal patch
LBP                              0.000            0.000           0.000            0.000           0.000                         0.053                         0.189
GLCM                             0.067            0.037           0.000            0.130           0.074                         0.033                         0.107
Gabor                            0.123            0.117           0.055            0.241           0.074                         0.086                         0.121
FFT fixed single-scale           0.147            0.123           0.145            0.148           0.148                         0.151                         0.004
FFT fixed multi-scale            0.233            0.196           0.327            0.185           0.185                         0.046                         0.025
FFT adaptive multi-scale         0.454            0.393           0.345            0.481           0.537                         0.026                         0.011
```

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IMAGE-LEVEL RESULTS (432 patches: 152 defect, 280 normal)
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                          Precision  Recall     F1  False-alarm rate  AUROC     AP    TP    FP     FN     TN
LBP                           0.211   0.053  0.084             0.107  0.643  0.430   8.0  30.0  144.0  250.0
GLCM                          0.348   0.105  0.162             0.107  0.656  0.467  16.0  30.0  136.0  250.0
Gabor                         0.534   0.204  0.295             0.096  0.786  0.572  31.0  27.0  121.0  253.0
FFT fixed single-scale        0.967   0.191  0.319             0.004  0.779  0.745  29.0   1.0  123.0  279.0
FFT fixed multi-scale         0.886   0.257  0.398             0.018  0.822  0.788  39.0   5.0  113.0  275.0
FFT adaptive multi-scale      0.959   0.467  0.628             0.011  0.830  0.792  71.0   3.0   81.0  277.0
```

Challenges Faced

- Making the comparison fair: The methods score on different scales and one could look better simply because the threshold used in that method was weaker. This was solved by normalizing all the scores for each method using the same defect-free data and all thresholds in the same manner.
There are few independent images. Patches from the same image are similar and not independent examples. The number of faulty source images is small, results are therefore uncertain. We dealt with this by re-sampling the source images and providing confidence intervals.

- Defects are small and rare – most defects are very small, and most patches have no defect at all, making pixel scores small and unstable. We have included defect-level and size-wise metrics to enable the detection to be not only based on pixel overlap.
Precision is dependent on data mix – the more defects and normal patches, the greater the precision. Next to it we report recall along with the false alarm rate and AUROC.

- Gabor filtering was about half a second/patch. We replaced it with a faster FFT-based one which provides the same results (the maximum difference is approximately 0.002 for values up to approximately 4000), and added a fast mode for testing.
Edge effects: The Fourier method produces artefacts at the edges of the patch, and only the interior of the patch (16 pixel margins) is scored. Any defects that are entirely within the margin are ignored, and the notebook prints the number of defects.

- Avoid over-tuning: The settings are set on DEV only and the TEST set is left unchanged for the final comparison.
The Fourier method did not detect defects that did not alter the repeatable pattern, only the brightness or the contrast, on real data and the detection of those defects decreased with contrast equalization (CLAHE) in real data preprocessing. Not confirmed yet on the real images of fabric.

Work To Be Done

Analyze the findings by fabric type. We will provide scores for each patch/ fabric (IoU, precision, recall and F1) and image-level detection for 7 fabrics, comparing the fixed and adaptive methods for each fabric. The adaptive method is expected to improve the results on fabrics of a clearly repeated weave.

- Examine the weave of each fabric. We will examine the acuity of the repeated pattern of each fabric and evaluate whether greater pattern acuity improves detection.
- Break down the results by defect type. We will examine the percentage of each twelve defect type that is detected and localized and identify easy and hard types. Care will be taken in reporting a very small number of images of a type.
- Analyze the results on the basis of defect size, small, medium and large, and compare the detection between the methods.

- Find the failure cases. The worst images will be examined (missed defects, false alarms, etc) and the reasons why (defects too small, defects not breaking the repeating pattern, edge effects, etc) will be discussed.

- If the numbers are small (particularly for fabrics and defect types), be sure to show confidence intervals to indicate the reliability of the numbers.
Deliverables: Fabric-wise and defect-wise performance tables and charts and a short discussion of failure cases.